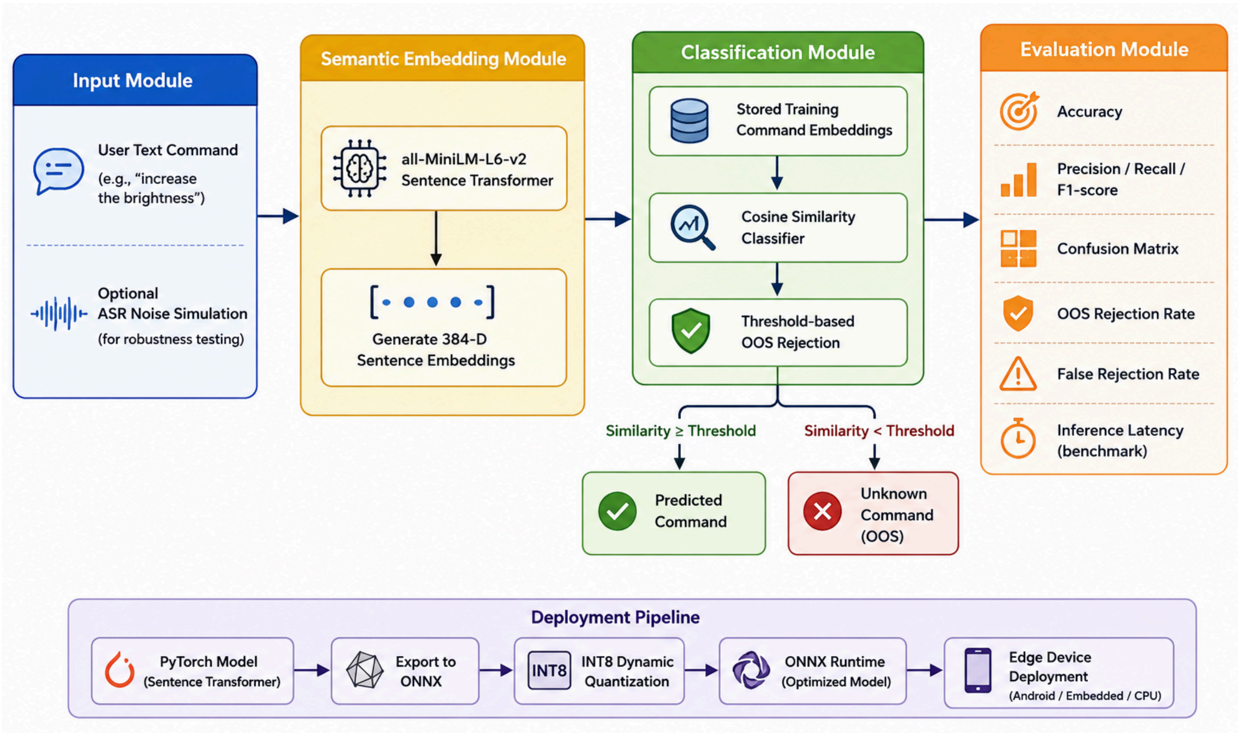
System Architecture:



The proposed system follows a modular pipeline for semantic speech command classification. The process begins with a user-provided text command, which may optionally undergo ASR-inspired noise simulation during evaluation to assess the model's robustness. The input is then passed to the all-MiniLM-L6-v2 Sentence Transformer, which converts the command into a 384-dimensional semantic embedding.

The generated embedding is compared with the stored training embeddings using a Cosine Similarity Classifier. Based on a predefined similarity threshold, the system either predicts the most semantically similar command or rejects the input as an Out-of-Scope (OOS) command if the similarity score is below the threshold.

The classifier is evaluated using standard performance metrics, including accuracy, precision, recall, F1-score, confusion matrix, OOS rejection rate, false rejection rate, and inference latency. For efficient deployment on resource-constrained devices, the pretrained PyTorch model is exported to the ONNX format and further optimized using INT8 dynamic quantization, resulting in a lightweight model suitable for edge deployment on platforms such as Android devices and embedded systems.